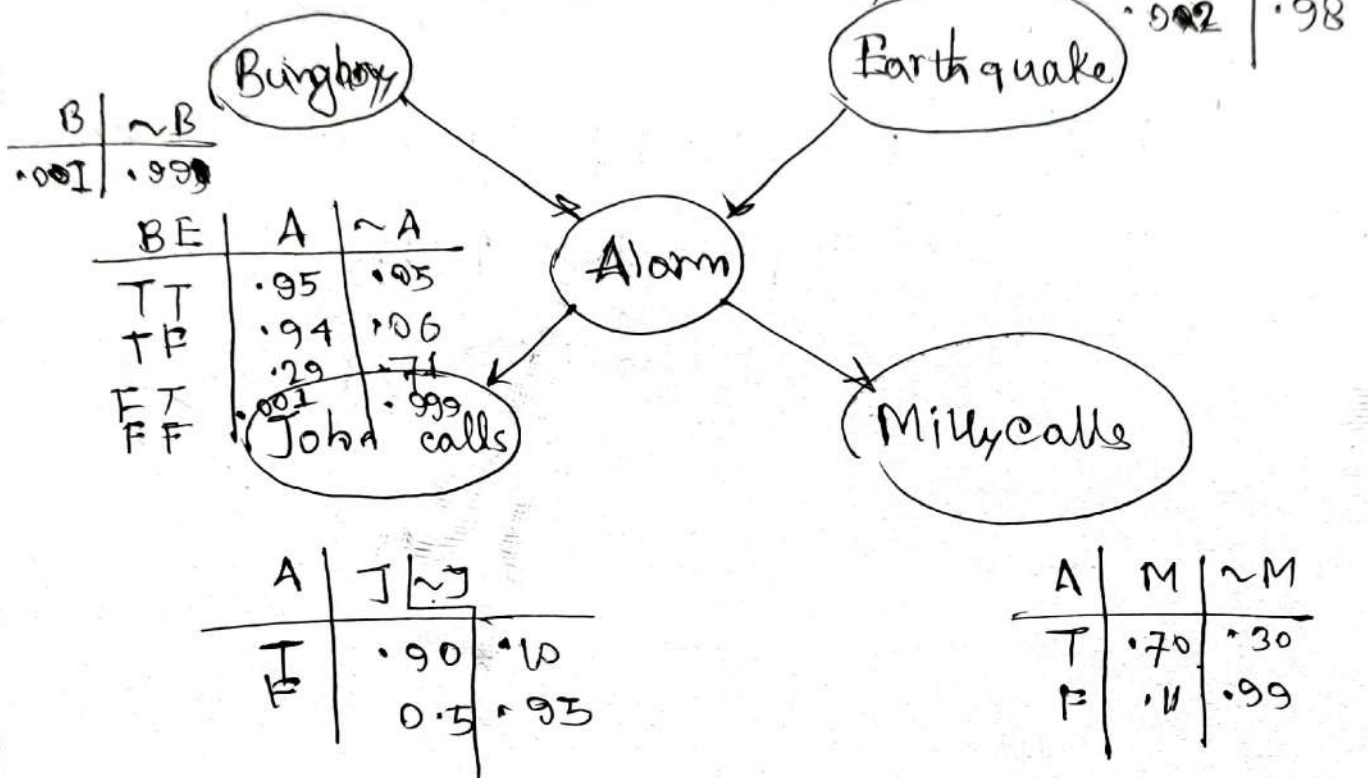


Joint Probability



What is the probability that John & Marry calls, given that alarm rang but no burglary and no earthquake happen?

$$\begin{aligned}
 & P(J, M, A, \sim B, \sim E) \\
 &= P(J|A) \times P(M|A) \times P(A|\sim B, \sim E) \times P(\sim B) \times P(\sim E) \\
 &\approx 0.00062
 \end{aligned}$$

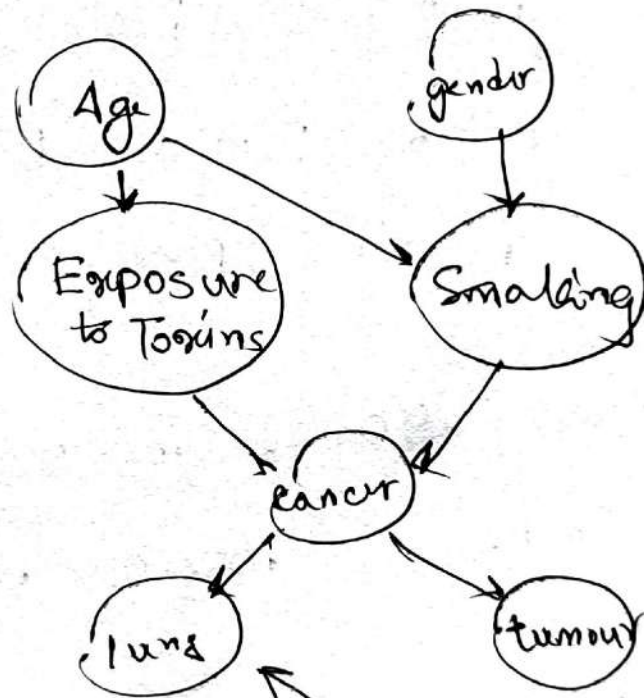
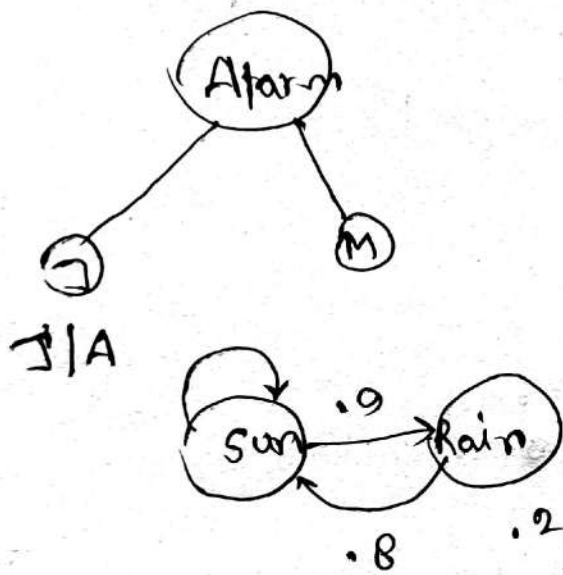
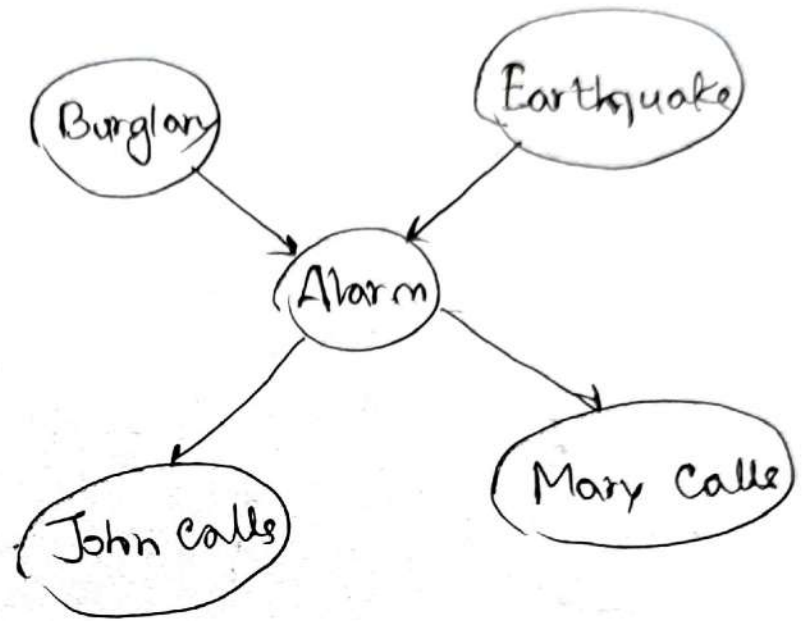
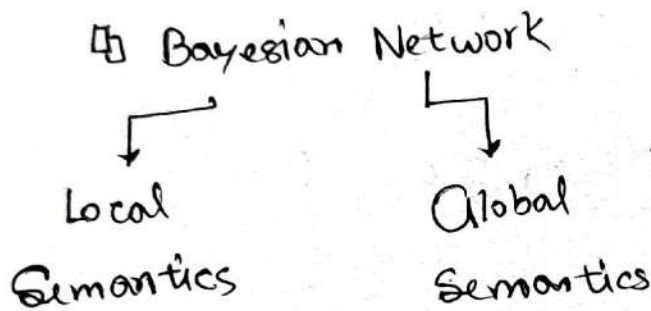
Marginal Probability

Probability of John calling?

$$\begin{aligned}
 &\Rightarrow P(J) \\
 &= P(J|A) P(A) + P(J|\sim A) P(\sim A)
 \end{aligned}$$

$$\begin{aligned}
 &= P(J|a) \left[p(a|b, e) p(b, e) + p(a|\sim b, e) p(\sim b, e) \right. \\
 &\quad \left. + p(a|b, \sim e) p(b, \sim e) + p(a|\sim b, \sim e) p(\sim b, \sim e) \right] \\
 &+ P(J|\sim a) \left[p(\sim a|b, e) p(b, e) + p(\sim a|\sim b, e) p(\sim b, e) \right. \\
 &\quad \left. + p(\sim a|b, \sim e) p(b, \sim e) \right. \\
 &\quad \left. + p(\sim a, \sim b, \sim e) p(\sim b, \sim e) \right]
 \end{aligned}$$

- Probabilistic Reasoning
- Bayesian Network
- Joint Probability
- Marginal Probability



- Markov Blanket
- Conditional probabilities in a network
- compact Conditional Distribution
- Conditional Independence

Shayan /

Inference (Reasoning) in Bayesian Networks

Q: What is the probability that, there is bulgary given that John & Mary calls?

$$= P(b|j,m) = P(j|a) P(m|a) P(a|b,e) P(b) P(e)$$

$$\Rightarrow \alpha \cdot P(b) \sum_a P(j|a) P(m|a) \sum_e P(a|b,e) P(e)$$

$$= \alpha \cdot P(b) \sum_a P(j|a) P(m|a) \{ P(a|b,e) P(e) + P(a|b,\neg e) P(\neg e) \}$$

$$= \alpha \cdot P(b) [P(j|a) P(m|a) \{ P(a|b,e) P(e) + P(a|b,\neg e) P(\neg e) \}$$

$$+ P(j|\neg a) P(m|\neg a) \{ P(\neg a|b,e) P(e) + P(\neg a|b,\neg e) P(\neg e) \}]$$

$$= \alpha \times 0.001 \times (90 \times 70 \times (0.95 \times 0.002 + 0.94 \times 0.98))$$

$$+ (0.05 \times 0.01 \times (0.05 \times 0.002 + 0.06 \times 0.98))$$

$$= \alpha \times 0.00059$$

$$P(\neg b|j,m)$$

$$\Rightarrow \alpha P(\neg b) \sum_a P(j|a) P(m|a) \sum_e P(a|\neg b,e) P(e)$$

$$= \alpha (P(\neg b) \sum_a (j|a) P(m|a) \{ P(a|\neg b,e) P(e) + P(a|\neg b,\neg e) P(\neg e) \})$$

$$= \alpha \cdot P(b) \left[P(j|a) P(m|a) \{ P(a|b, e) P(e) + P(a|b, \neg e) P(\neg e) \} \right]$$

$$= \alpha \cdot P(b) \left[P(j|a) P(m|a) \{ P(a|b, e) P(e) + P(a|b, \neg e) P(\neg e) \} \right. \\ \left. P(j|\neg a) P(m|\neg a) \{ P(\neg a|b, e) P(e) + P(\neg a|b, \neg e) P(\neg e) \} \right]$$

$$= \alpha \times 0.99 \left[0.90 \times 0.70 (0.29 \times 0.002 + 0.09 \times 0.98) \right] + \\ \left[0.10 \times 0.30 \times (0.71 \times 0.002) + 0.99 \times 0.98 \right]$$

$$P(b|j, m) = \frac{\alpha \times 0.0015}{\alpha}$$

$$\alpha = \frac{1}{P(b|j, m) + P(\neg b|j, m)} = \frac{1}{0.0059 + 0.0015} = 478.5$$

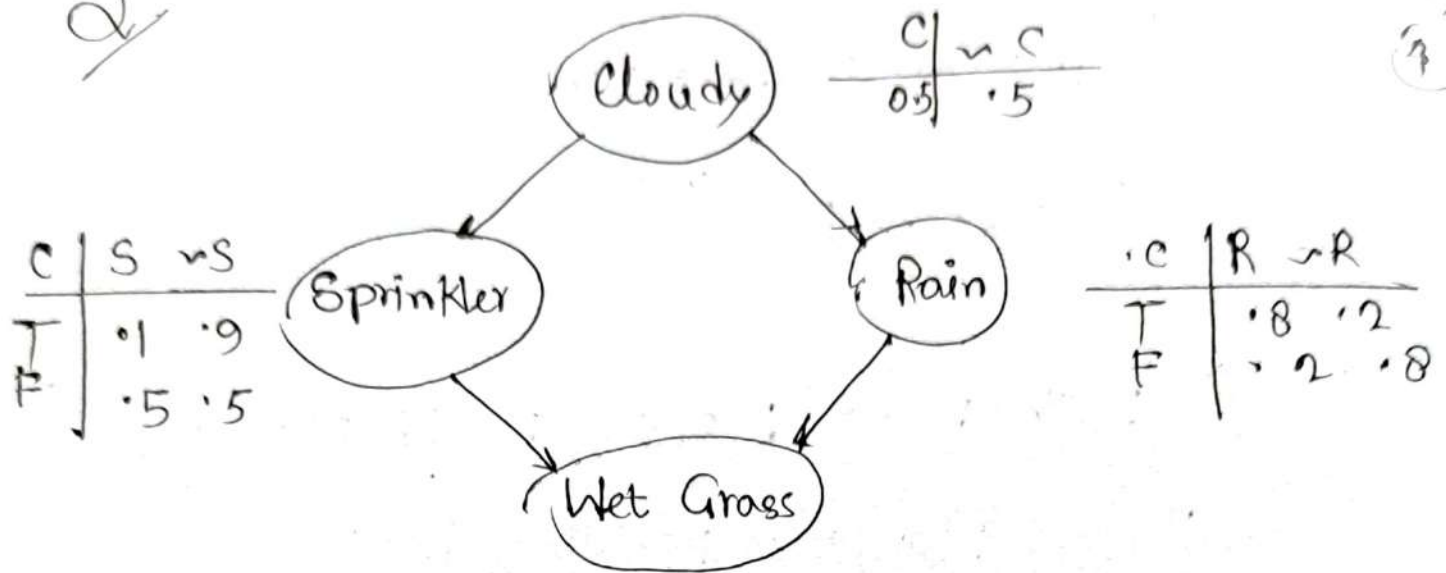
$$P(b|j, m) = 0.0059 \times 478.5 = 0.28$$

$$P(\neg b|j, m) = 0.0015 \times 478.5 = 0.72$$

$$0.28 + 0.72 = 1$$

2

(1) has



Q: Given that grass is wet, what's probability that it's due to rain and not the sprinkler assuming a cloudy day?

$$P\left(\frac{R, \sim S}{A} \middle| \frac{W, C}{B}\right)$$

$$= \frac{P(R, \sim S \cap W, C)}{P(W, C)}$$

$$\Rightarrow P(R, \sim S | W, C)$$

$$= \frac{P(R, \sim S, W, C)}{P(W, C)}$$

$$= \frac{P(W|R, \sim S) \times P(R|C) \times P(\sim S|C) \times P(C)}{P(W) \times P(C)}$$

$$= \frac{.90 \times .80 \times .90}{P(W)}$$

$$= \frac{.90 \times .80 \times .90}{P(W|S, R) \times P(S|C) \times P(R|C) + P(W|S, \sim R) \times P(S|C) \times P(\sim R|C) + P(W|\sim S, R) \times P(\sim S|C) \times P(R|C) + P(W|\sim S, \sim R) \times P(\sim S|C) \times P(\sim R|C)}$$

$$= .0648$$

$$\frac{(.99 \times .10 \times .80) + (.90 \times .10 \times .20) + (.90 \times .90 \times .80) + (.00)}{.079}$$

$$= \frac{.0648}{.079}$$

$$= 0.87$$